

AY25 PROGRAM ASSESSMENT REPORT

v2026.1

Tracking Data

CIM code (e.g., ME-CM)	PhD-MSE
Program Name	PhD in Materials Science and Engineering
Assessment Year	2025
Program Assessment Coordinator	Heng Wang
Year of Current Assessment Plan	2025
No Assessment This Year per Plan	☒
LO(s) Assessed	LO1, LO2 (collecting and archiving data only)
Presented at program authority meeting date: <i>Please attach minutes</i>	TBD

Methodology

LO#1

- ∄ Learning goal assessed: Graduates will be able to apply proper computational and numerical techniques in contexts relevant to Materials Science and Engineering.
- ∄ Sample size: All students completing MMAE 501
- ∄ Semesters in which artifacts were collected: Fall, Spring
- ∄ Artifact: For each course, performance of the students in 4 questions chosen from a pool of questions specifically designed to assess this learning goal. The questions are delivered to the students in the last third of the semester. Each student will be scored out of 4.
- ∄ Proficiency ratings
 - ⌞ scores > 75% points indicate **P** proficient.
 - ⌞ scores > 55% points indicate **M** meets expectations.
 - ⌞ scores < 55% points indicate **D** does not meet expectations.

LO#2

- ∄ Learning goal assessed: Graduates will have the ability to conduct research, publish scientific findings, and disseminate that knowledge.
- ∄ Sample size: All students completing their comprehensive exam and thesis defense.
- ∄ Semesters in which artifacts were collected: Fall, Spring
- ∄ Artifact: A particularly designed questionnaire filled out by the thesis committee after the comprehensive and defense exams. The committee members come to a consensus and rate the students on a scale of 1-5 on each of the following:
 - ⌞ Research problem is clearly described

- ⌘ Literature and context of the work are demonstrated
- ⌘ Solution approach, methods and/or tools are described
- ⌘ Results are analyzed, interpreted and communicated well
- ⌘ Written and oral communication are effective
- ⌘ Critical thinking consistent with an independent researcher
- ⌘ A record of publication is established (on thesis defense only)
- ⌘ Proficiency ratings
 - ⌘ scores of 4 points and above for all items indicate **P** proficient.
 - ⌘ scores of 3 points and above for all items indicate **M** meets expectations.
 - ⌘ scores below 3 points on any item indicate **D** does not meet expectations.

Results

Artifact samples

Artifact for LO1 from MMAE 501. Four questions from a bank of 10 were selected and assigned as a quiz. Sample scores listed below. "C" for correct, "I" for incorrect.

1
Multiple Choice
1 point

Given a system of linear differential equations $\frac{dx}{dt} = Ax$, where A is a square matrix, which statement is true about the general solution?

- ☒ If A has repeated eigenvalues, the solution may involve terms of the form $t^k \cdot e^{\lambda t}$
- ☐ It always contains n linearly independent solutions, where n is the size of A
- ☐ It can be expressed as $x(t) = e^{(At)x(0)}$ only if A is invertible
- ☐ The stability of the system depends solely on the determinant of A

MS-MAE LO1 MS-ASAR LO1

3
Multiple Choice
1 point

Consider the Sturm-Liouville problem: $-(p(x)y')' + q(x)y = \lambda r(x)y$ with appropriate boundary conditions. Which statement is correct?

- ☒ The eigenfunctions form an orthogonal set with respect to the weight function $r(x)$
- ☐ The eigenvalues are always real, regardless of the functions $p(x)$, $q(x)$, and $r(x)$
- ☐ The number of nodes in the n th eigenfunction is always exactly n
- ☐ The lowest eigenvalue is always positive if $q(x) > 0$ for all x

MS-MAE LO1 MS-ASAR LO1

2
Multiple Choice
1 point

In the calculus of variations, what is the primary difference between constrained and unconstrained optimization problems?

- ☒ Constrained problems require the use of Lagrange multipliers
- ☐ Constrained problems always lead to simpler Euler-Lagrange equations
- ☐ Unconstrained problems can only be solved using direct methods
- ☐ The extremals for constrained problems are always smooth curves

MS-MAE LO1 MS-ASAR LO1

4
Multiple Choice
1 point

For a complex function $f(z) = u(x,y) + iv(x,y)$, which condition ensures that $f(z)$ is analytic?

- ☒ $u_x = v_y$ and $u_y = -v_x$ (where u_x is the partial derivative of u w.r.t x etc.)
- ☐ $u_x = v_y$ and $u_y = v_x$
- ☐ $u^2 + v^2 = \text{constant}$
- ☐ $u_{xx} + u_{yy} = 0$

MS-MAE LO1 MS-ASAR LO1

	Sample 1		
P	C, C, C, C (100%)		
M			
D			

LO#2 artifact. No PhD-MSE student took comprehensive exam in F24-S25.

	Sample 1			
Research problem is clearly described				
Literature and context of the work are demonstrated				
Solution approach, methods and/or tools are described				

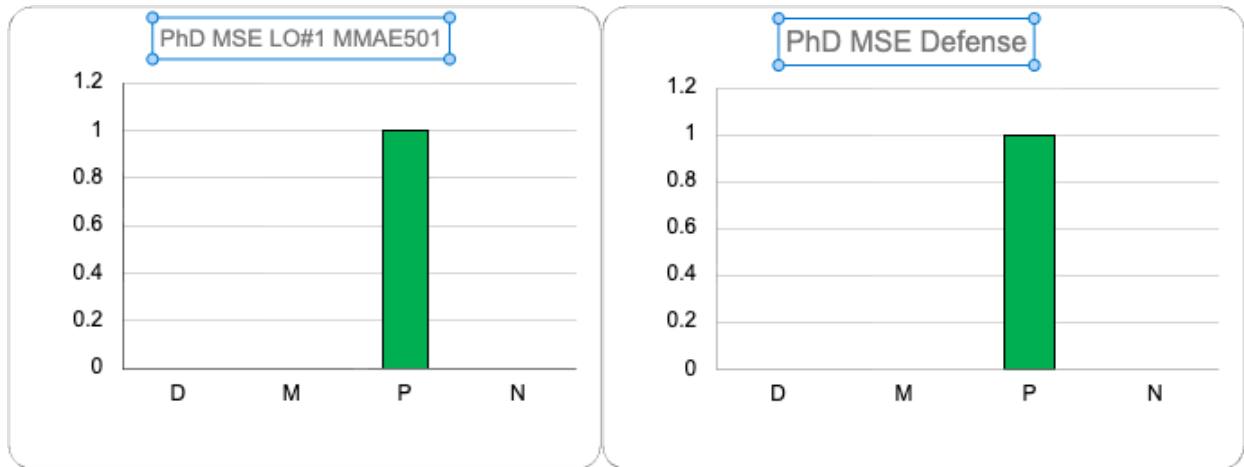
Results are analyzed, interpreted and communicated well				
Written and oral communication are effective				
Critical thinking consistent with an independent researcher				

One PhD-MSE student took thesis defense in F24-S25. Analysis will be performed when a sufficient amount of sample has been collected.

	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5
Research problem is clearly described	5				
Literature and context of the work are demonstrated	5				
Solution approach, methods and/or tools are described	5				
Results are analyzed, interpreted and communicated well	5				
Written and oral communication are effective	5				
Critical thinking consistent with an independent researcher	5				
A record of publication is established	5				

Data table

Charts below show the distribution of results from different courses. Y axis is the number of students.



Description of findings

There are no sufficient samples to perform meaningful analysis. The department decided to do so every 4-5 years when more data points are collected.

Continuous Improvement, CI, plans

Prior CI plan impact

Also, please refer to the previous assessment report for the LOs assessed. Were any CI plans recommended? Were they implemented? Did the plans work (were improvements seen in this cycle)?

☒ no previous CI plans available

This cycle's CI plan

Describe any CI plans developed from this cycle's data, with

- Specific curricular or instructional changes in response to this assessment data,
- A timeline for implementation,
- As well a description of who will participate in and manage these plans

☒ After consideration and presentation to the program authority, no CI plans are recommended.